

```

1  rooms = {'A': 'Dirty', 'B': 'Dirty'}
    # initial environment
2  location = 'A'
    # vacuum starts in room A
3
4  step = 1
5  while True:
6      print(f"\nStep {step}:")
7      print("Vacuum Location:", location)
8      print("Room Status:", rooms)
9
10     # Action: Clean if dirty
11     if rooms[location] == "Dirty":
12         print(f"Action: Suck dirt in
                Room {location}")
13         rooms[location] = "Clean"
14     else:
15         print(f"Action: Room {location}
                already clean")
16
17     # Check goal condition
18     if rooms['A'] == "Clean" and
        rooms['B'] == "Clean":
19         print("\nAll Rooms Clean ✅")
20         break
21
22     # Action: Move to the other room
23     if location == 'A':
24         print("Action: Move Right")

```

Run

```
6     print(f"\nStep {step}:")
7     print("Vacuum Location:", location)
8     print("Room Status:", rooms)
9
10    # Action: Clean if dirty
11    if rooms[location] == "Dirty":
12        print(f"Action: Suck dirt in
13              Room {location}")
14        rooms[location] = "Clean"
15    else:
16        print(f"Action: Room {location}
17              already clean")
18
19    # Check goal condition
20    if rooms['A'] == "Clean" and
21        rooms['B'] == "Clean":
22        print("\nAll Rooms Clean ✅")
23        break
24
25    # Action: Move to the other room
26    if location == 'A':
27        print("Action: Move Right →
28              Room B")
29        location = 'B'
30    else:
31        print("Action: Move Left → Room
32              A")
33        location = 'A'
```

Run

Step 1:

Vacuum Location: A

Room Status: {'A': 'Dirty', 'B': 'Dirty'}

Action: Suck dirt in Room A

Action: Move Right → Room B

Step 2:

Vacuum Location: B

Room Status: {'A': 'Clean', 'B': 'Dirty'}

Action: Suck dirt in Room B

All Rooms Clean 

=== Code Execution Successful ===